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Week 6 Lab - CS010A

Fare Calculation

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Problem Overview

1. Given commuter's start time, calculate the toll fare amount.
2. Input: User's start time in 24-hour format: hours and minutes.
3. Output: Amount of money to be paid by user.
4. You will be given a table. You must compare the input hours and minutes with the table's conditions and decide the amount.
5. Example:
   a. Input: 13 24
   b. Output: 6.9
6. You have to implement a **function** that returns this fare.
7. This function will be called from the main function of your program.

Time	Fare
00:00 - 06:59	$6.15
07:00 - 09:59	$7.95
10:00 - 14:29	$6.90
14:30 - 19:59	$8.95
20:00 - 23:59	$6.40

Primer on functions

Example - 1:

1. Ordering coffee at Coffee Bean and Tea Leaf.
    a. You give:
        i. Type: “Latte”
        ii. Temp: “Hot”
        iii. Size: 12 oz
        iv. Money: $6.00
    b. You get:
        i. Cup of coffee

Observe: You did not prepare the coffee. You just gave the inputs. The shop prepare gave you the coffee.

Primer on functions

Example - 2:

1. Asking ChatGPT to explain the assignment.
    a. You give:
        i. Prompt: “I’m stuck since yesterday night… Why is this loop not working?? I rechecked every logic and ran this test like 50 times and still this one test case is missing… Why is life so hard.., am I just larping lowkey… but I know deep inside I am good… Please please please help me…”
        ii. Attachments: “Screenshot”
        iii. Type: “GPT 6.1 Sol”
        iv. Effort: “Extra High”
    b. You get:
        i. “You forgot user input in 1st line.”

Observe: You used ChatGPT as a tool. You just gave the inputs. ChatGPT gave you the answer.

Primer on functions

In Example-1: Coffee Bean and Tea Leaf shop played the role of a function.

In Example-2: ChatGPT played the role of a function.

In either case,

- You “used” the function to finish a task.
- You “supplied” some inputs to the function.
- You “received” some outputs from the function.
- But, you never did the job of the function.

This same logic can be translated into C++ code.

Primer on functions - in code

Format:

```
int calculate(int a, int b, char op) {
    int ans = 0;

    if (op == 'A') {
        ans = a + b;
    }
    else if (op == 'S') {
        ans = a - b;
    }
    return ans;
}
```

Primer on functions - in code

Format:

```
int calculate(int a, int b, char op) {
    int ans = 0;

    if (op == 'A') {
        ans = a + b;
    }
    else if (op == 'S') {
        ans = a - b;
    }
    return ans;
}
```

Matches output data type at the end.

Return data type.

Function name.

No spaces.
A verb – represents action.

Parameters.

Inputs. Comma separated.
With data type.

Return statement.

Output. Return keyword.
Exactly 1 Return value.

Primer on functions - in code

```
int main() {

   // vars
   int x;
   int y;
   int z;
   char c;

   // inputs
   cin >> x >> y >> c;

   // function call
   int z = calculate(x, y, c);

   // outputs
   cout << z << endl;

   return 0;
}
```



This is “function calling”.

You pass the inputs here.

Then, the program **jumps** to the function to execute its statements.

Once the function returns, the program **jumps back** to the main function.

Recap of today’s problem

- Input: Hours and Minutes in 24-hr format.
- Output: Fare amount to pay.
- Use given table to compare the inputs and decide output.
- Use function to implement this feature.
- Function has 4 main parts:
  - Return type.
  - No-space Function name.
  - Comma-separated parameters with data type.
  - Return statement with one variable.
- Do not use early return.
- Do no cout inside the function.
- Stick to content taught in yesterday’s class and zyBooks.
- Code Plan → Test Cases → Code